

## **GENE 8160: ADVANCED EVOLUTIONARY GENETICS**

Spring 2011, 3 credits

**Class meetings:** Monday and Wednesday 9:05-11:00 am, Life Sciences Room B116

**Prerequisite:** GENE 8150 or GENE 6000

**Instructors:** Kelly Dyer and John Wares

### **Contact Information and Office Hours**

KD: Email: [kdyer@uga.edu](mailto:kdyer@uga.edu), Phone: 706-542-3154, Office: Life Sciences C226

Office hours: Monday and Wednesday 11:15 – 12 and by appointment (please email)

JW: Email [jpwares@uga.edu](mailto:jpwares@uga.edu), Phone: 706-542-7720, Office: Life Sciences C326

Office hours: By appointment (please email)

### **Course description and goals**

This course explores the process of evolution at the molecular level. We will use basic evolutionary principles to explore how DNA changes through time; to infer population and species history from DNA sequences; to determine what forces have shaped the evolution of genes and genomes; and to understand the relationship between molecular evolution and phenotypic evolution. This field is large and diverse, and combines population genetics, classical genetics, phylogenetics, molecular biology, and genomics. The goals of this class are to introduce you to some of the main questions, give you a feel for the kinds of data used, and expose you to the tools that are used to answer questions.

### **Texts and assigned readings**

Reading for this course will be primarily articles and reviews from the primary literature. Articles will be available as PDF files on eLearning Commons.

You will want to keep the following book handy to use as a basic reference:

*Principles of Population Genetics*, 4th edition (2008) by D. L. Hartl and A. G. Clark.

### **Other useful resources**

*Molecular Ecology*, J. Freeland. Wiley.

*Population Genetics, A concise guide*, 2<sup>nd</sup> ed. J. Gillespie. Johns Hopkins Press

*Fundamentals of Molecular Evolution*, 2<sup>nd</sup> ed. By D. Grauer and W.-H. Li. Sinauer Press.

*Computational Molecular Evolution*, Z. Yang. Oxford University Press.

### **Class format, Participation, and Attendance**

This class will combine lectures, discussions, and computer activities. We expect that you come to class prepared and ready to engage in whatever activities we will be doing that day.

Participation is essential for lively discussion, and your contribution to this is reflected in the participation portion of your grade. Regular attendance in lecture is expected and necessary for learning the material, which is cumulative in nature. After two unexcused absences, each additional absence will result in the loss of 1% from your final average, except in the case of a documented medical or extreme personal emergency.

### **Grading**

Grades are A-F and will be based on three exams (20% each), assignments and presentations (30%), and participation (10%).

### **Lecture Handouts**

A pdf copy of the lecture slides will be available before class on eLearning Commons.

**Exams**

Three non-cumulative exams will be given during lecture, on 2/14/11, 3/28/11, and 5/2/11. Makeup exams will only be given in the case of a documented medical or extreme personal emergency. All make-up exams will be oral examinations. If you submit a request for a regrade, your entire exam will be regraded.

**Short assignments**

Short assignments may come in many flavors, depending on the nature of the material. In addition to traditional problem sets, throughout the semester you will write several 'response' papers (~ 1 page typed and in essay format) to journal articles or class discussions. For journal articles, these should contain (1) the main question is being addressed, (2) what the authors did, (3) what the authors found, (4) the significance of the findings, and (5) a question you had about the paper (e.g. What left you baffled? What techniques were new? Were the conclusions inappropriate? Was the design inadequate?) and (6) two to three questions or topics for discussion.

**Discussions**

Frequently we will spend part of a class period discussing a paper from the primary literature. When you present a paper, you will introduce the paper in a 10-15 minute presentation and then lead the discussion of the paper. You will likely want to come prepared with some 'kick off' questions for discussion. You will be evaluated based on if you introduced the reading in a knowledgeable manner, presented data/figures/results clearly and concisely, engaged the class in insightful discussion, effectively answered questions, and in general helped to explain the science. You need to know the paper you are presenting backward and forward and inside out, and you probably want to read related papers as well.

**Electronics**

Please turn your phone off during class and office hours.

**Religious Holidays**

If you plan to miss a class because of a religious holiday, you must talk to one of the instructors at least fourteen days prior to the class to make arrangements to make-up missed work.

**Honor Code**

All students will abide by the UGA Student Honor Code: "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others." All suspected incidences of academic dishonesty, as outlined under prohibited conduct in the UGA academic honesty policy (<http://www.uga.edu/ovpi/honesty/sect05.htm>), will be treated following UGA guidelines ([http://www.uga.edu/ovpi/honesty/culture\\_honesty.htm](http://www.uga.edu/ovpi/honesty/culture_honesty.htm)).

**Disclaimer**

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.

## GENE 8160: Tentative course schedule

| Class | Date   |    | Tentative Topic                   |                              |
|-------|--------|----|-----------------------------------|------------------------------|
| 1     | M 1/10 | JW | Life history I                    | Life History, Trade-offs     |
| 2     | W 1/12 | JW | Life history II                   | Biogeography, Drift          |
|       | M 1/17 |    | No class: MLK Holiday             |                              |
| 3     | W 1/19 | JW | Effective population size I       | Drift, Effective Pop Size    |
| 4     | M 1/24 | JW | Effective population size II      | Diversity, Life History      |
| 5     | W 1/26 | JW | Molecular population genetics I   | Model testing                |
| 6     | M 1/31 | JW | Molecular population genetics II  | Molecular pop gen            |
| 7     | W 2/2  | JW | Molecular population genetics III | Population structure         |
| 8     | M 2/7  | JW | Phylogeography I                  | Phylogeography               |
| 9     | W 2/9  | JW | Phylogeography II                 | Phylo diversity              |
| 10    | M 2/14 | JW | EXAM #1                           |                              |
| 11    | W 2/16 | KD | Phylogenetic methods I            | Intro to phylogenetics       |
| 12    | M 2/21 | KD | Phylogenetic methods II           | Maximum Likelihood           |
| 13    | W 2/23 | KD | Phylogenetic methods III          | Bayesian methods             |
| 14    | M 2/28 | KD | Phylogenetic methods IV           | Rates and dates              |
| 15    | W 3/2  | KD | Phylogenetic methods V            | Gene trees vs. species trees |
| 16    | M 3/7  | KD | Phylogenetic methods VI           | Comparative method           |
| 17    | W 3/9  | KD | Phylogenetic methods VII          | Selection along a phylogeny  |
|       | M 3/14 |    | No class: Spring Break            |                              |
|       | W 3/16 |    | No class: Spring Break            |                              |
| 18    | M 3/21 | KD | Phylogenetic methods VIII         | Case studies                 |
| 19    | W 3/23 | KD | Phylogenetic methods IX           | Case studies                 |
| 20    | M 3/28 | KD | EXAM #2                           |                              |
| 21    | W 3/30 | KD | Evolutionary genomics I           | Intro to population genomics |
| 22    | M 4/4  | KD | Evolutionary genomics II          | Candidate locus tests        |
| 23    | W 4/6  | KD | Evolutionary genomics III         | Genome scans                 |
| 24    | M 4/11 | KD | Evolutionary genomics IV          | Noncoding DNA evolution      |
| 25    | W 4/13 | KD | Evolutionary genomics V           | Gene regulatory evolution    |
| 26    | M 4/18 | KD | Evolutionary genomics VI          | Duplications                 |
| 27    | W 4/20 | KD | Evolutionary genomics VII         | Genome evolution             |
| 28    | M 2/25 | KD | Evolutionary genomics VIII        | Case studies                 |
| 29    | W 4/27 | KD | Evolutionary genomics IX          | Case studies                 |
| 30    | M 5/2  | KD | EXAM #3                           |                              |